

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: tCGtoPkt_Gen4 Update

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Increasing the value of tCGtoPkt Gen4 from 550ns to 700ns for Host Routers
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Benefits as a result of the changes:

The existing maximal bound for tCGtoPkt_Gen4 is defined to match specific implementations, the change is aimed to provide more margin for higher latency power-optimized Host Router designs
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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NA

An analysis of the hardware implications:
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NA

An analysis of the software implications:
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NA

An analysis of the compliance testing implications:
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Need to update test 5.104 in the USB4 Protocol CTS
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Actual Change

(a). Section 5.5

From Text:

Parameter	Description	Min	Max	Units
tCGtoPkt_Gen4	The maximum time between receiving a Credit Grant Packet and sending a Tunneled Packet, assuming the Tunneled Packet is available and was originally not sent due to a lack of credits. This parameter only applies to a Gen 4 Link.	--	550	ns

To Text:

Parameter	Description	Min	Max	Units
tCGtoPkt_Gen4	The maximum time between receiving a Credit Grant Packet and sending a Tunneled Packet, assuming the Tunneled Packet is available and was originally not sent due to a lack of credits. This parameter only applies to a Gen 4 Link while the link is not in the middle of link transitions. Note: It is recommended for a Host Router not to exceed 550nSec	--	550 (Device Router) 700 (Host Router)	ns

(b). Section 5.3.2.1.3

From Text:

In order to assure minimum bandwidth, the round trip delay between sending a Credit Grant Packet to getting the next Tunneled Packet needs to be bounded. If an Egress Adapter is not transmitting Tunneled Packets due to a lack of credits, a Router shall send a Tunneled Packet for a Path when the Egress Adapter receives a Credit Grant Record for the Path and the Credit Grant Record contains an advanced PCA. Unless another packet or Ordered Set is transmitted in the interim, the Tunneled Packet shall be sent within tCGtoPkt_Gen3 (for a Gen 3 Link) or tCGtoPkt_Gen4 (for a Gen 4 Link) time after receiving the Credit Grant Packet. The time is measured on the USB Type-C connector from the first received bit of the Credit Grant Record to the first transmitted bit of the Tunnel Packet.

To Text:

In order to assure minimum bandwidth, the round trip delay between sending a Credit Grant Packet to getting the next Tunneled Packet needs to be bounded. If an Egress Adapter is not transmitting Tunneled Packets due to a lack of credits, a Router shall send a Tunneled Packet for a Path when the Egress Adapter receives a Credit Grant Record for the Path and the Credit Grant Record contains an advanced PCA. Unless another packet or Ordered Set is transmitted in the interim **or the USB4 Link is in the middle of a Link Transition**, the Tunneled Packet shall be sent within tCGtoPkt_Gen3 (for a Gen 3 Link) or tCGtoPkt_Gen4 (for a Gen 4 Link) time after receiving the Credit Grant Packet. The time is measured on the USB Type-C connector from the first received bit of the Credit Grant Record to the first transmitted bit of the Tunnel Packet.